

# Impacts of Public Early Childhood Education in Chile on Maternal Fertility and Labor Market Outcomes

Short-run effects — progress update

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# Where we are

## What we do

**Question:** does opening a public preschool nearby change mothers' labor supply and fertility?

**Variation:** 447 neighborhood units (UVs) open their first JUNJI/Integra preschool in 2015–2024 vs. 4,145 that never do

**Design:** staggered event study at the UV level, Sun–Abraham estimator (Baker et al., *JEL* 2026)

**Data:** woman-level panel linking administrative registries — RSH/FPS, SII/AFP earnings, Civil Registry births, preschool enrollment

## What we find so far

**First stage:** enrollment **+2pp** (33% → 35%), from the opening year, no pre-trends; strongest at ages 2–3

**2015 cohort** (~40% of treated) shows almost no effect; without it, **≈+3pp** (preliminary)

**Fertility:** no effect so far — a fairly precise zero

**Formal employment:** **+1pp** from year 4 on; pre-trends not significant but rising

**Earnings** (mothers who work): **suggestive +2–4%**, not yet significant

**Months worked:** no effect so far

**Cleaner control group:** same results

## Project status

**Done:** data, treatment, balance, first stage, first labor and fertility estimates

**In progress:** 2015 cohort check, earnings including non-working mothers

**Next:** pre-trend robustness, mechanisms, writing

# Research question

## Question

What are the effects of the public expansion of early childhood education in Chile on mothers' labor market participation and fertility?

## What we know

**Mixed evidence:** positive effects on maternal employment in Quebec and Argentina, close to zero in Norway

Recent Latin American evidence comes from **lotteries and experiments** (Rio de Janeiro, Nicaragua)

Less is known about a **national, decade-long public expansion** observed in administrative data

## Our contribution

**First study to use neighborhood-unit (UV) variation:** preschools geolocated to the UV and linked to individual administrative records

**Fertility** as a second margin, alongside labor supply

**What is a UV?** *Unidad Vecinal*: the official sub-municipal territorial unit — ~6,900 UVs vs. 346 municipalities.

⇒ **Finer geography: a sharper measure of exposure and more variation to identify the effect.**

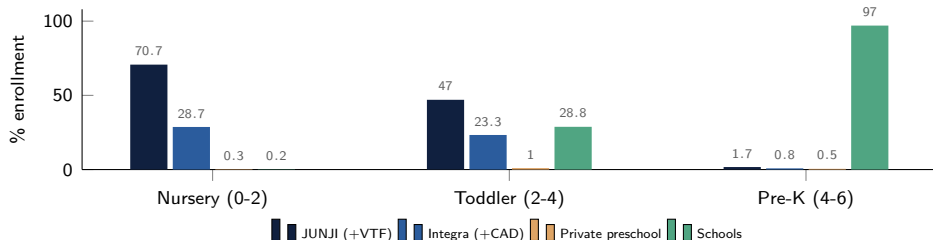
Quebec: Baker et al. (2008); Argentina: Berlinski & Galiani (2007); Norway: Havnes & Mogstad (2011); Rio: Attanasio et al. (2025); Nicaragua: Hojman & López Bóo (2022).

# Public provision expanded sharply between 2015 and 2024

Mixed provision: **public** (JUNJI, Integra, municipal/SLEP) vs. **private** (subsidized/paid)

**700+ new public preschools** opened between 2015 and 2024

Net attendance rate (ages 0–5) rose from **50% to 55%** over the period, despite pandemic-era setbacks



Source: authors' elaboration based on Figura 5, Matrícula Oficial 2023, Centro de Estudios MINEDUC.

⇒ **Ages 0–4: JUNJI and Integra are the main providers — our focus.**

# Identification: staggered-adoption event study

**Design:** staggered-adoption difference-in-differences with event study

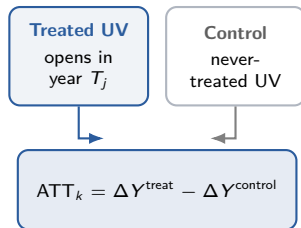
**Unit of treatment:** the UV, the year it opens its first public preschool

**Unit of analysis:**  $UV \times year$  panel, built from a woman  $\times$  year panel linked to each woman's UV of residence

**Fixed effects:**  $UV + year$  (baseline); woman FE on the woman  $\times$  year panel as a **robustness check**

**Estimand:** ATT of UV-level treatment (Baker et al., *JEL* 2026)  
— an **intent-to-treat effect of access**, not individual take-up

⇒ **We compare changes in UVs that get their first preschool vs. UVs that never do — an intent-to-treat effect of access.**



# A UV is treated when it opens its first preschool

## Treatment definition

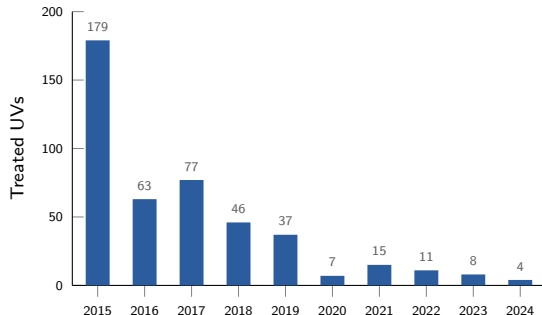
A UV is **treated** the year it opens its first public preschool (JUNJI/Integra) — goes from 0 to 1+ centers.

| Group                 | Definition                                    | N (UVs)      | %           |
|-----------------------|-----------------------------------------------|--------------|-------------|
| Excluded              | Already had 1+ center in 2014 (out of design) | 2,265        | 33.0%       |
| Never-treated         | 0 in 2014, stays at 0 through 2024            | 4,145        | 60.5%       |
| Treated               | 0 in 2014, opens first center 2015–2024       | 447          | 6.5%        |
| <b>Total universe</b> |                                               | <b>6,857</b> | <b>100%</b> |

Source: base\_cobertura\_cp.dta. Design sample (muestra\_t1==1) = Never-treated + Treated = 4,592 UVs.

⇒ We identify the effect of the first preschool (0 → 1): 447 treated vs. 4,145 never-treated.

## Treatment cohorts by year



**2015 cohort is the largest:** 179 UVs (40%), tied to the Bachelet preschool expansion

**71% open in 2015–2017;** only 45 UVs open after 2019

**2014 as base year** lets us identify the 2015 cohort as  $0 \rightarrow 1$ ; without it, treated UVs drop from 447 to  $\sim 268$

But the 2015 cohort has **only one pre-period** and sits at the edge of the panel  $\rightarrow$  we come back to it below

Source: base\_cobertura\_cp.dta. 447 treated UVs (0 centers in 2014).

$\Rightarrow$  **Most openings are early (2015–17): long post-treatment windows for most treated UVs.**

# Empirical specification and estimator

$$Y_{ut} = \lambda_u + \lambda_t + \gamma \overline{\text{Age}}_{ut} + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - G_u = k\} + \varepsilon_{ut}$$

$Y_{ut}$ : average outcome of women in UV  $u$ , year  $t$  (built from the woman  $\times$  year panel).  $\lambda_u, \lambda_t$ : UV and year fixed effects; SE clustered by UV.

$\overline{\text{Age}}_{ut}$ : average age of mothers in the UV-year (labor market specification).

$\beta_k$ : effect  $k$  years from opening, relative to  $k = -1$  (omitted);  $k \leq -5$  binned.

## Estimation strategy

**Baseline TWFE** as reference — but with staggered treatment, TWFE can use already-treated units as controls (Goodman-Bacon problem). Robust estimator: **Sun–Abraham** (`eventstudyinteract`), never-treated UVs as control group. Framework: Baker, Callaway, Cunningham, Goodman-Bacon & Sant’Anna (*JEL*, 2026).

**Main threat**: non-random placement of openings (JUNJI/Integra prioritizing UVs by care demand or socioeconomic profile). **UV FE** absorb permanent differences across UVs; **pre-period coefficients** ( $k < -1$ ) test whether trends were already diverging.

⇒ **Main estimator: Sun–Abraham with never-treated UVs as controls; TWFE as reference.**

## Data: what we have and how each sample is built

| Sample                      | Source data                                               | Key variables                                       |
|-----------------------------|-----------------------------------------------------------|-----------------------------------------------------|
| Coverage (UV $\times$ year) | JUNJI/Integra transparency requests + UV shapefile (MDSF) | Number of centers, treatment status, opening cohort |
| Women panel (base)          | RSH (2016–2024) + FPS (2014–2015)                         | UV of residence, UV population, child density       |
| Labor market                | + Rentas (SII/AFC) + Cotizaciones (AFP)                   | Annual income, months worked, employment            |
| Fertility                   | + Civil Registry birth records (via RIS)                  | Birth in the year, children under 5                 |
| First stage (enrollment)    | + Matrícula parvularia (child-level)                      | Enrollment, enrollment rate                         |

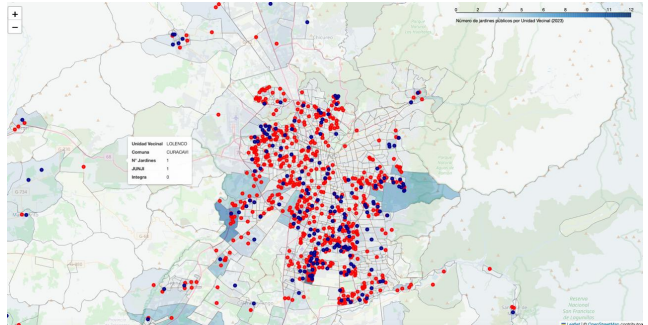
⇒ All administrative registries, linked at the individual level — no survey self-reports.

# Every preschool is matched to its 2024 neighborhood unit

Each JUNJI/Integra preschool is **geolocated** and matched to the UV where it sits (spatial join)

UV boundaries **changed a lot** over time  
→ we traced them and mapped everything to **2024 boundaries**

Preschools and women's UV of residence share **one common geography** across all years

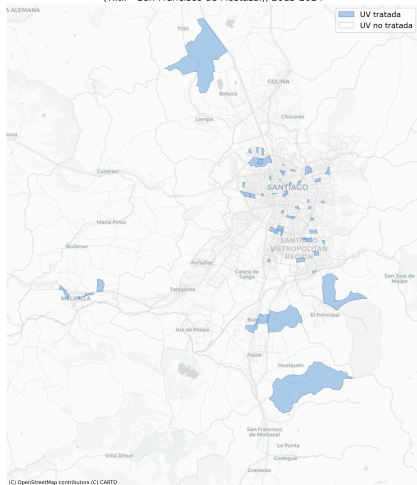


Región Metropolitana. ● JUNJI ● Integra Shading: public preschools per UV. Source: base\_cobertura\_cp.dta.

⇒ **One common geography: every preschool and every woman mapped to 2024 UV boundaries.**

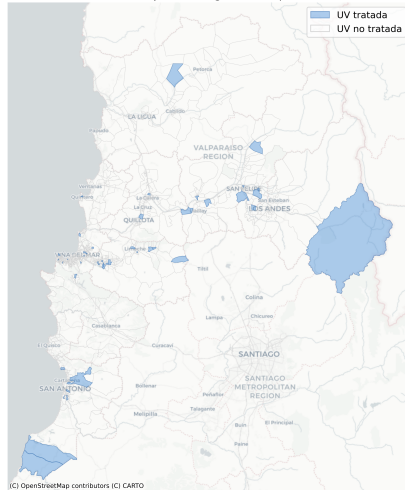
## Región Metropolitana

UVs tratadas ( $0 \rightarrow \geq 1$  jardín), Gran Santiago  
(Tiltil - San Francisco de Mostazal), 2015-2024



## Valparaíso

UVs tratadas ( $0 \rightarrow \geq 1$  jardín), Región de Valparaíso, 2015-2024



Illustrative examples. Shaded: treated UVs ( $0 \rightarrow 1+$  preschools), 2015–2024.

# Who is in each sample, and what we measure

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## 1. **Samples** (selected at the individual level)

**Fertility:** women of fertile age (18–49)

**Labor market:** women with a child  $< 5$  *that specific year* — not “ever had a child”

**First stage:** children of treatment-sample mothers, birth to birth+4

## 2. **Outcomes** (averaged at $UV \times \text{year}$ )

**Fertility:** share of women who had a child that year (Civil Registry birth records)

**Labor market:** average annual income, months worked, share employed, dependent vs. independent

**First stage:** enrollment rate

⇒ **Each outcome is measured on its relevant population, from administrative records, averaged at the  $UV \times \text{year}$  level.**

## Treated and never-treated UVs look alike — except in size

**Labor market sample** (mothers with a child <5) — covariate levels in 2014, before any opening

| Variable              | Unweighted  |         |              | Weighted by UV population |         |              | Sig. <sup>a</sup> |
|-----------------------|-------------|---------|--------------|---------------------------|---------|--------------|-------------------|
|                       | Never-tr.   | Treated | Norm. diff.  | Never-tr.                 | Treated | Norm. diff.  |                   |
| Annual income (M CLP) | 3.60        | 3.55    | -0.033       | 3.95                      | 3.89    | -0.039       | *                 |
| Months worked         | 8.14        | 8.10    | -0.024       | 8.44                      | 8.37    | -0.058       | *                 |
| Age                   | 29.24       | 29.25   | 0.008        | 29.26                     | 29.29   | 0.026        |                   |
| N. children <5        | 1.11        | 1.11    | 0.045        | 1.11                      | 1.11    | 0.049        |                   |
| Hours worked          | 37.07       | 37.68   | 0.099        | 36.55                     | 36.81   | 0.068        |                   |
| Works (share)         | 0.45        | 0.45    | 0.023        | 0.48                      | 0.48    | -0.006       |                   |
| Education (code)      | 4.98        | 5.02    | 0.028        | 5.20                      | 5.18    | -0.020       |                   |
| <b>UV population</b>  | 1126.79     | 2011.08 | <b>0.410</b> | 3530.92                   | 5287.26 | <b>0.340</b> | ***               |
| Child density 0–5 UV  | 0.06        | 0.06    | 0.155        | 0.06                      | 0.06    | 0.139        | ***               |
| N (UVs)               | 3,909 / 422 |         |              | 3,909 / 422               |         |              |                   |

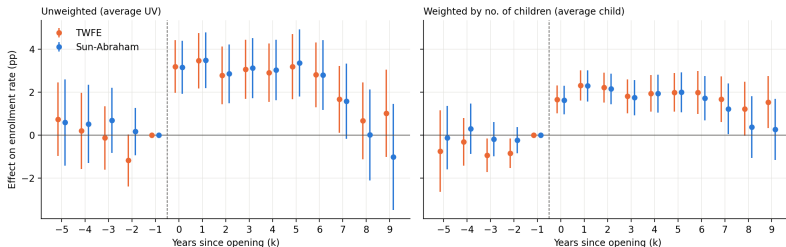
<sup>a</sup> t-test at the woman level (\*  $p < 0.05$ , \*\*\*  $p < 0.01$ ), for reference only; normalized differences prioritized (concern threshold: 0.25).

The UV population gap is **expected**: the State prioritizes new preschools where need and population are larger. Same pattern in the fertility sample (App. A4).

⇒ **Selection is in *where* preschools open (UV size), not in *who* lives there → UV fixed effects.**

# First stage: enrollment rises the year a preschool opens and stays higher

First stage: enrollment event study (95% CI, ref.  $k = -1$ )



**Weighted (main):**  $\approx +2\text{pp}$  on a 33% base ( $\approx +6\%$ ), immediate and stable for  $k = 0-6$

Unweighted:  $\approx +3\text{pp}$  — larger UVs move less with the same new places

**No pre-trends:** small and not significant

$k = 7-9$ : only early cohorts (mostly 2015) observed  
→ composition, not a fading effect

► Average effects (A6)

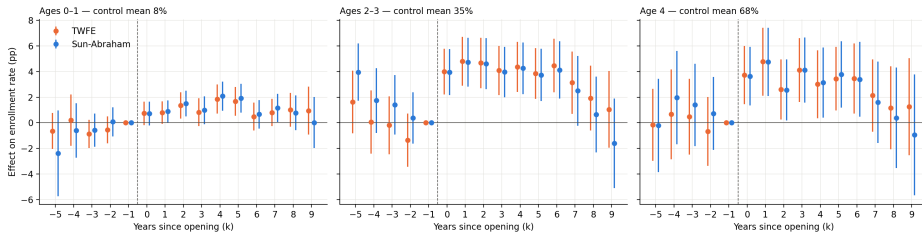
⇒ **A new preschool raises enrollment from 33% to 35%: about 3 more children enrolled per UV.**

Sun-Abraham and TWFE, never-treated UVs as controls, UV and year FE, 95% CI clustered by UV, reference  $k = -1$ . 14 / 33

Children aged 0-4.

# First stage by child age: strongest at ages 2–3

First stage by child age (95% CI, ref.  $k = -1$ )



| Child age       | Control mean | SA $k = 0-6$  | Relative |
|-----------------|--------------|---------------|----------|
| Ages 0-1        | 8%           | +1.3pp        | +15%     |
| <b>Ages 2-3</b> | 35%          | <b>+4.2pp</b> | +12%     |
| Age 4           | 68%          | +3.6pp        | +5%      |

**Ages 2-3:** largest effect, immediate and stable

**Ages 0-1:** small in pp but largest in relative terms; grows over time

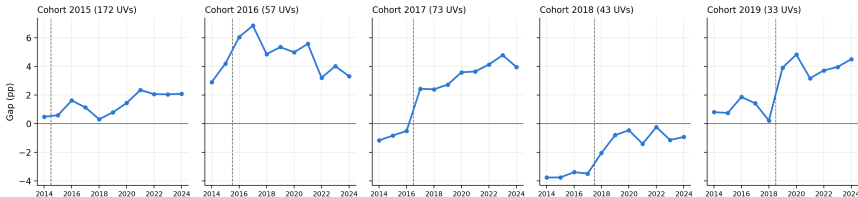
**Age 4:** also positive

**Caution:** ages are measured at different points in time across datasets (to check)

⇒ **Enrollment rises at all ages, most at ages 2-3.**

# An open issue: the 2015 cohort

Enrollment gap = (% of children 0-4 enrolled in the cohort UVs) – (% in never-treated UVs), each year, in pp.  
Gap 0 = same rate as controls. Effect = how much the gap changes after the opening (dashed line) relative to before.



**172 UVs** — 41% of treated UVs, ~half of the sample at  $k = 6$ , the only cohort at  $k = 9$

Its effect is **very small** ( $\approx 0.7$ pp in raw gaps) and it has only one pre-period (2014)

**Without it:**  $\approx +3$ pp vs. 1.9pp, ~50% larger and stable over time (preliminary)

**Hypothesis:** registration error at the edge of the panel — preschools that already operated in 2014 but are recorded as 2015 openings.

**Next:** carefully review the coverage data provided by JUNJI/Integra.

⇒ **The 2015 cohort pulls the average effect down — possibly a data error to check.**

# A cleaner control group: never-treated UVs can be exposed too

**Concern:** children from a never-treated UV can enroll in a new preschool next door → control enrollment rises → **effect underestimated**

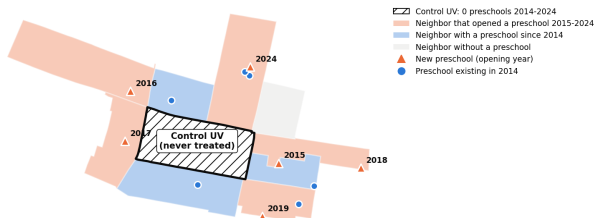
We built a **neighbor map**: for every UV, which adjacent UVs had or opened a preschool (shared border, 2024 boundaries)

Example: this control UV has 9 neighbors  
— **5 open a preschool** during the period

⇒ **Goal:** a larger first stage, comparing treated UVs with controls that could *not* use a neighbor's new preschool.

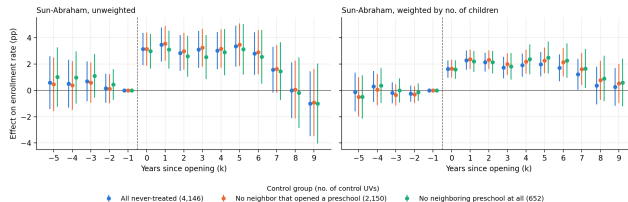
\* Neighbors only capture proximity to home. If mothers use preschools near their workplace, spillovers may reach UVs that are not adjacent.

A 'never-treated' UV in La Cisterna  
5 of its 9 neighboring UVs open a preschool in the period



# A cleaner control group gives the same first stage

First stage with different control groups (95% CI, ref.  $k = -1$ )



| Control group                                 | UVs          | Median area               | Control mean | SA weighted, $k = 0-6$ |
|-----------------------------------------------|--------------|---------------------------|--------------|------------------------|
| All never-treated (main)                      | 4,146        | 2.8 km <sup>2</sup>       | 33.2%        | 1.92                   |
| <b>No neighbor opened a preschool 2015–24</b> | <b>2,150</b> | 3.8 km <sup>2</sup>       | 32.7%        | <b>2.14</b>            |
| No neighboring preschool in any year          | 652          | <b>6.7 km<sup>2</sup></b> | 28.2%        | 2.13                   |
| <i>Treated UVs</i>                            | 422          | 1.5 km <sup>2</sup>       |              |                        |

Same effect with all three groups and clean pre-trends. The strictest group is not a good comparison: its UVs are  $\sim 4\times$  larger than treated ones (isolated areas)  $\rightarrow$  preferred clean control: **no neighbor opened**.

$\Rightarrow$  **Spillovers to controls do not explain the small first stage.**

# Why is the first stage small?

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## What we checked

**2015 cohort:** very small effect, possibly a data error → without it,  $\approx 3\text{pp}$

**Cleaner control group:** same  $\approx 2\text{pp}$  → spillovers to controls are not the reason

## What our outcome measures

Whether children **living** in the treated UV are enrolled in **any** preschool

## A new preschool can fill up without moving our outcome

It raises enrollment only if it takes local children who were **not enrolled before**

**Substitution:** local children already enrolled elsewhere switch to the new preschool → no net change

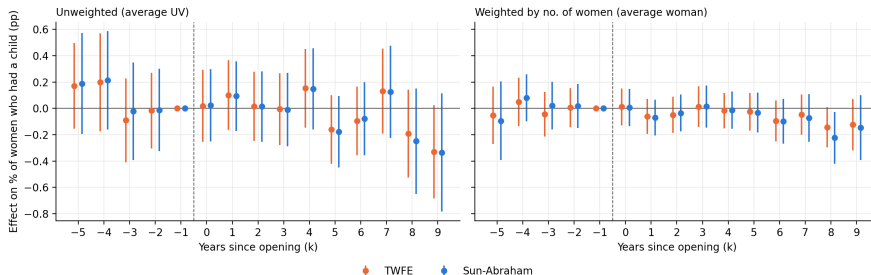
**Mothers use other preschools**, e.g. near their workplace in another UV → new places go to children from other UVs

**Future work:** enrollment records show which preschool each child attends → who fills the new preschool, where they live, and which preschools local mothers use.

⇒ Enrollment of local children can move little, even if the new preschool fills up.

# Fertility: no effect yet

Fertility: event study of had a child (95% CI, ref.  $k = -1$ ). Control mean = 4.3%



Treated and control UVs decline together; **no jump at opening**

Clean pre-trends

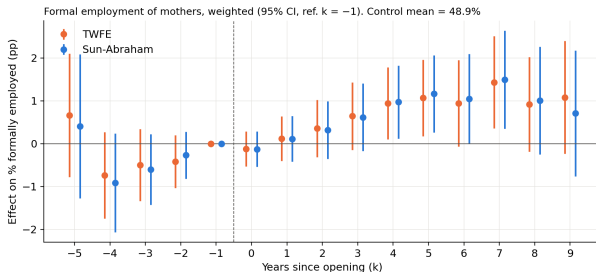
A **fairly precise zero**: rules out large effects

Still checking: cumulative outcome (having had a child)

⇒ **Expected: having a child depends on many factors, and a nearby preschool might weigh little.**

Women aged 18–49. Outcome: share who had a child in the year (control mean 4.3%). Controls for average age.

# Mothers' formal employment rises by 1pp from the fourth year



Nothing at opening; **significant every year from  $k = 4$  to 7**

## Why gradual?

Each year's sample = mothers with a child  $< 5$

At  $k = 0$ , many children are already 3–4: little time to use the preschool

From  $k \approx 4$ , every mother in the sample had it available **since her child was born**  
→ exposure builds up

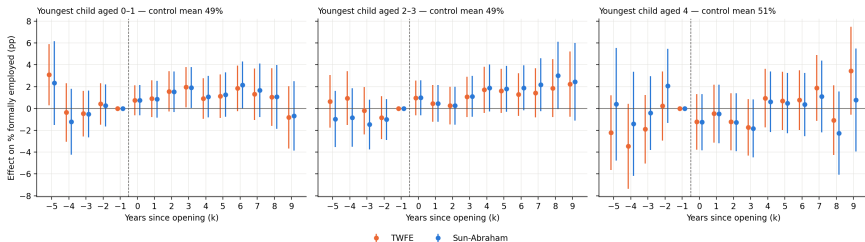
**Caution:** pre-trends not significant (joint test  $p = 0.82$ ) but slightly rising; pre and post years come from different cohorts.

⇒ **A new preschool raises mothers' formal employment by  $\approx 1$ pp from the fourth year.**

Weighted by number of mothers. Mothers with a child under 5. Employed = any taxable income or social security contribution in the year.

# Employment effect appears for mothers whose youngest child is 0–3

Formal employment of mothers by age of youngest child (unweighted, 95% CI, ref. k = -1)



Youngest child **0–1 and 2–3**: positive, +1 to +2pp

Youngest child **4**: no effect

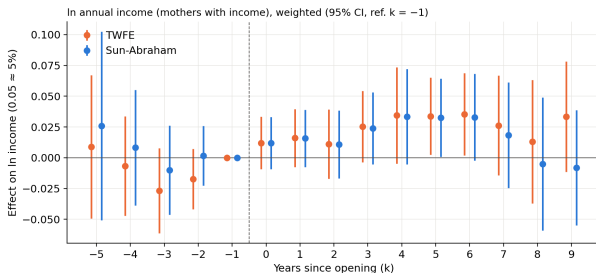
Fits the mechanism: mothers of younger children face a tighter care constraint; at age 4, most children are already enrolled (68%)

Not significant one by one — the **pattern across groups** is what is informative

⇒ **The effect comes from mothers of children aged 0–3, who need care the most.**

Unweighted (weighted version not yet estimated).

## Earnings: suggestive increase of $\approx 2\text{--}3\%$



Same **gradual shape** as employment; clean pre-trends

$\approx +3\%$  at  $k = 4\text{--}6$ ; average  $k = 0\text{--}6$ :

$+2.3\% \approx$  CLP 10–20k per month, not yet significant

### Still to do

In earnings is measured **only among mothers with earnings** → new entrants may pull it down

Add earnings **in levels**, including zeros for mothers without earnings

**Months worked** (App. A5): no effect so far, but also conditional on working → re-estimate with 0 for mothers without work

⇒ Mothers who work earn  $\approx 2\text{--}3\%$  more from the fourth year: a suggestive result.

## Next steps

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### Immediate

**How many mothers move across UVs?**

2015 cohort: review the coverage data provided by JUNJI/Integra

Re-estimate all outcomes **without the 2015 cohort**

Earnings in levels, including zeros

Months worked **including zeros**

Fertility: **cumulative** outcome (having had a child)

### Short term

Employment: balanced event study (same cohorts pre and post)

Child age: align age definitions across datasets

### Medium term

Mechanisms: care constraints (RSH self-reports, App. A7)

Who fills the new preschool? (child-level enrollment)

Write results section

# Appendix

## Appendix A1. Full identification assumptions and placebo check

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**Parallel trends:** assessed via pre-opening event-study coefficients; should be statistically indistinguishable from zero

**No anticipation:** mothers do not adjust decisions before the effective opening — depends on how much advance notice openings receive (to analyze)

**Exogeneity of timing:** opening timing should not correlate with unobserved UV characteristics. **Main threat:** JUNJI/Integra may prioritize UVs with higher care demand or specific socioeconomic profiles

### Placebo check

An analogous event study will be estimated using the **pre-treatment birth rate** of the UV as the outcome. Non-significant pre-treatment coefficients would strengthen the parallel-trends assumption of the main model.

## Appendix A2. Data limitation: UV assignment in 2014–2015

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FPS (2014–2015) has no reliable UV field → those two years are anchored to the woman's **2016 UV** as a proxy

**Consequence:** women who left the panel before 2016 are dropped from 2014–2015 — precisely the earliest pre-trend years

RIS has not delivered a better historical crosswalk despite repeated requests; only unstandardized UV name strings are available

**Idea, not yet tested:** fuzzy matching on UV name strings

## Appendix A3. Task status

| Block              | Task                                              | Status      | Note                                                                   |
|--------------------|---------------------------------------------------|-------------|------------------------------------------------------------------------|
| 2. Data generation | Coverage base (preschools $\times$ UV, 2014–2024) | ready       | Verified; private preschools (transparency) still a possible extension |
|                    | Women panel (2014–2024, incl. FPS anchor)         | ready       | 2014–2015 anchored to 2016 UV (see A2)                                 |
|                    | Rentas / Cotizaciones cleaning                    | advanced    | Independent/state/FF.AA. cotizaciones still pending                    |
|                    | Children registry cleaning                        | ready       | tuvo_hijo, n_hijos_menor5                                              |
|                    | Merge panel + outcome variables                   | ready       | By rut_inn                                                             |
| 3. Panel analysis  | Treated / non-treated definition                  | ready       | grupo_t1, g1, muestra_t1                                               |
|                    | First-stage estimation (enrollment)               | ready       | +3pp; 2015 cohort under review                                         |
| 4. Data analysis   | Balance / validity check                          | ready       | UV-level balance table                                                 |
|                    | Methodology choice                                | ready       | Sun–Abraham within Baker et al. (JEL 2026) framework                   |
|                    | Treatment definition                              | ready       | Extensive margin, 0 $\rightarrow$ 1                                    |
|                    | Run regressions (labor, fertility)                | ready       | Re-run without 2015 cohort pending                                     |
| 5. Mechanisms      | Analyze results                                   | in progress | Pre-trend robustness for employment                                    |
|                    | Heterogeneity by group / level                    | in progress | By child age (first stage, employment)                                 |
| 6. Writing         | Public-policy implications                        | not started |                                                                        |
|                    | —                                                 | in progress |                                                                        |

## Appendix A4. Balance in the fertility sample

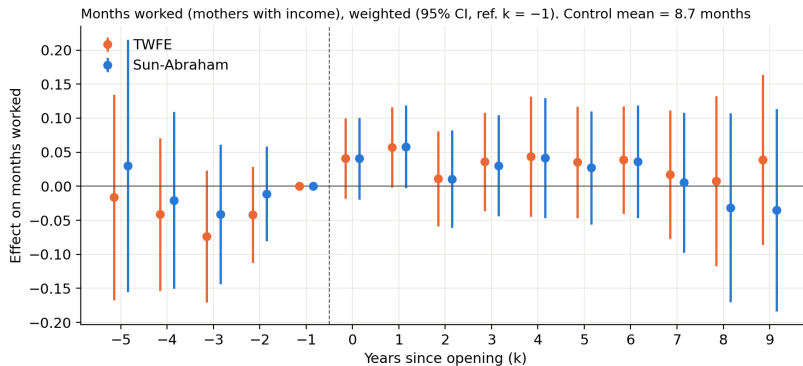
**Fertility sample** (women aged 18–49) — covariate levels in 2014, before any opening

| Variable                      | Unweighted  |         |              | Weighted by UV population |         |              | Sig. <sup>a</sup> |
|-------------------------------|-------------|---------|--------------|---------------------------|---------|--------------|-------------------|
|                               | Never-tr.   | Treated | Norm. diff.  | Never-tr.                 | Treated | Norm. diff.  |                   |
| Birth rate                    | 0.057       | 0.058   | 0.031        | 0.06                      | 0.060   | 0.051        |                   |
| Age                           | 31.90       | 31.83   | -0.053       | 31.74                     | 31.82   | 0.095        |                   |
| Children <5 per woman         | 0.303       | 0.319   | 0.156        | 0.31                      | 0.320   | 0.117        | **                |
| <b>UV population</b>          | 1111.42     | 2025.96 | <b>0.423</b> | 3540.63                   | 5305.76 | <b>0.342</b> | ***               |
| Child density 0–5 UV          | 0.0590      | 0.0631  | 0.176        | 0.06                      | 0.064   | 0.148        | **                |
| <b>Women in sample per UV</b> | 233.60      | 445.45  | <b>0.401</b> | 795.29                    | 1234.46 | <b>0.347</b> | ***               |
| N (UVs)                       | 3,973 / 422 |         |              | 3,973 / 422               |         |              |                   |

<sup>a</sup> t-test at the UV level (\*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ ), for reference only; normalized differences prioritized (concern threshold: 0.25). UVs with at least one sample woman in 2014, hence more never-treated UVs than in the labor sample.

⇒ **Same conclusion as the labor sample: women look alike, UVs differ in size.**

## Appendix A5. Months worked: no effect so far



Weighted. Mothers with earnings: about +0.03 months on a base of 8.7, clean pre-trends.

**Caveat:** currently conditional on working. Next: months worked with 0 for mothers without work.

## Appendix A6. First stage: average effect is $\approx 2$ pp for the average child

[◀ Back to first stage](#)

| Sun–Abraham average (pp)          | Weighted (main) | Unweighted  |
|-----------------------------------|-----------------|-------------|
| <b>Post, <math>k = 0-6</math></b> | <b>1.92</b>     | 3.11 (0.58) |
| Post, $k = 0-9$                   | 1.53            | 2.23 (0.63) |
| Pre, $k = -5--2$                  | -0.06           | 0.49 (0.59) |
| Joint pre-test                    | —               | $p = 0.90$  |
| Control mean                      | 33.2%           | 32.4%       |
| Relative effect ( $k = 0-6$ )     | <b>+6%</b>      | +10%        |

**Weighted** = effect for the average child (weights: children aged 0–4 in the UV). **Unweighted** = effect for the average UV.

**What changed since last meeting:** the model now includes *all* event-time dummies. The negative  $k = -2$  we flagged came from TWFE; it disappears with Sun–Abraham.

Average of event-study coefficients; SE (in parentheses) uses their full covariance matrix. Weighted SE pending.  $k = 0-6$  is our main window (balanced across cohorts).

⇒ **A clear first stage: 33% → 35%, immediate, no pre-trends — but small.**

## Appendix A7. Self-reported outcomes (RSH): exploratory

| Outcome (mothers, RSH)                    | Control mean | Weighted                       | Unweighted      |
|-------------------------------------------|--------------|--------------------------------|-----------------|
| <b>Not looking for work: no childcare</b> | 11.8%        | −0.4pp                         | − <b>0.7pp*</b> |
| Looking for work                          | 18.6%        | +0.6pp                         | +0.8pp          |
| Weekly hours worked                       | 36.5 h       | +0.1 h                         | −0.3 h          |
| Not looking for work: household chores    | 12.0%        | not interpretable (pre-trends) |                 |

**Most interesting:** fewer mothers say they don't look for work because they have no one to care for their children (−6% relative), from  $k = 1$ , clean pre-trends — the most direct measure of the care-constraint mechanism

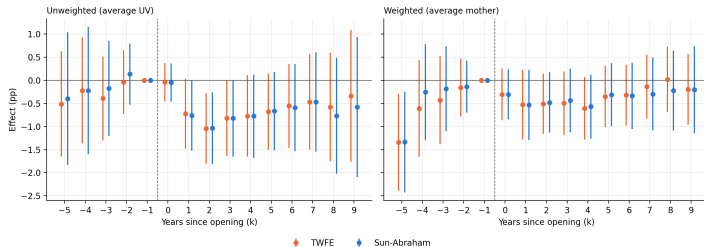
Weighted: same direction, weaker

**Caveats:** self-reported; the RSH form is not updated every year, and families may update it when applying for benefits (e.g., a preschool) → timing may depend on treatment

Sun–Abraham average effect,  $k = 0$ –6. \*  $p < 0.05$ . Weighted SE pending.

## Appendix A8. RSH: fewer mothers lack childcare

Not looking for work because "no one to leave the children with" (% of all mothers) — control mean 11.8% (95% CI, ref.  $k = -1$ )



Fewer mothers say they don't look for work because they have no one to leave the children with. Unweighted:  $-0.7\text{pp}$  ( $-6\%$ ), significant from  $k = 1$ , clean pre-trends; weighted: same direction, weaker

**Caveats** (exploratory): only mothers who neither work nor look for work answer the reason; all others are coded 0, so it is a % of all mothers — if more mothers start working, it falls too (part of the effect). The RSH form is not updated every year: the same answer can repeat for several years, so changes are slow and noisy. Some UVs have very few answers ( $\sim 4$  women).